



Mechanical Engineerng

Lecture No. 2

Forces analysis

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Sources:

- 1- R. C. Hibbeler. Engineering mechanics statics. Fourteen edition. 2016.
- 2- R. C. Hibbeler. Engineering mechanics dynamics. Fourteen edition. 2016.

Force:- may be defined as any action that tends to change the state of rest or motion of a body to which it is applied .Force is the **vector quantity**. Sometimes it is necessary to resolve a force into two components in order to study its pulling and pushing effect in two specific directions. For example, in the below Fig. , **F** is to be resolved into two components along two axis **x, y** .

$$\mathbf{F} = \mathbf{F}_x + \mathbf{F}_y$$

where $\mathbf{F}_x$ and $\mathbf{F}_y$ are *vector components* of $\mathbf{F}$ in the x and y directions.

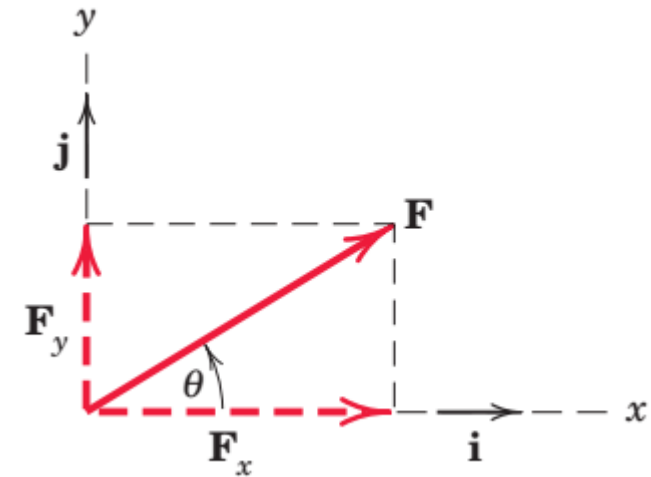
$$\mathbf{F} = F_x \mathbf{i} + F_y \mathbf{j}$$

where the scalars F_x and F_y are the x and y *scalar components* of the vector $\mathbf{F}$:

$$F_x = F \cos \theta$$

$$F_y = F \sin \theta$$

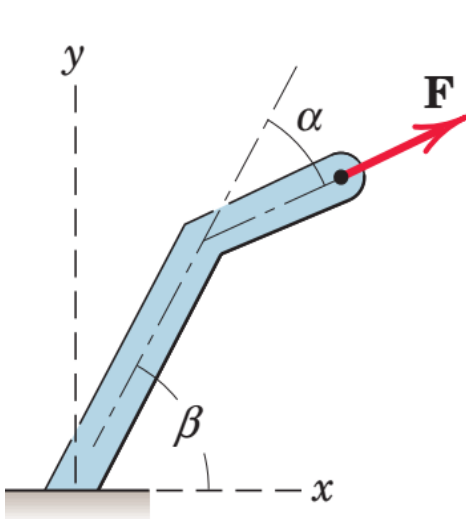
$$\theta = \tan^{-1} \frac{F_y}{F_x}$$



$\mathbf{i}$ and $\mathbf{j}$ are unit vectors in x and y axis.

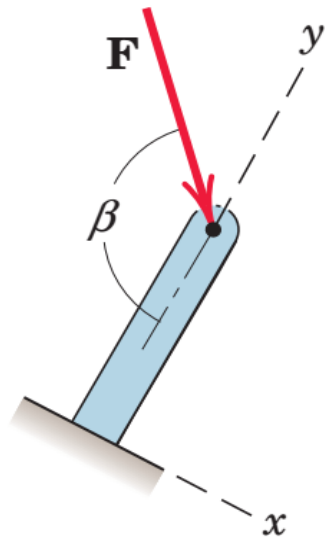
$$F = \sqrt{F_x^2 + F_y^2}$$

The axes are not always given in **horizontal and vertical directions**, and angles need not be measured **CCW(Counter Clockwise)** from the x-axis, and the **origin of coordinates** need not be on the line of action of a force. Therefore, it is essential that we be able to determine the correct components of force no matter how the axes are oriented or how the angles are measured. below Fig. suggests a few typical examples of vector resolution in two components **x,y**:



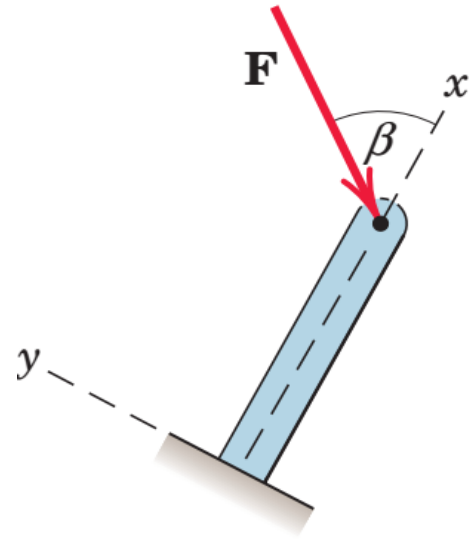
$$F_x = F \cos(\beta - \alpha)$$

$$F_y = F \sin(\beta - \alpha)$$



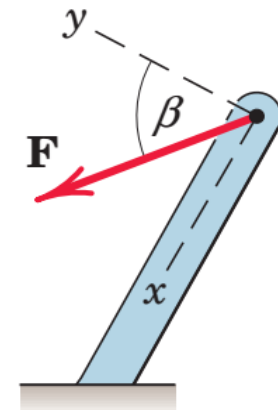
$$F_x = F \sin(\pi - \beta)$$

$$F_y = -F \cos(\pi - \beta)$$



$$F_x = -F \cos \beta$$

$$F_y = -F \sin \beta$$



$$F_x = F \sin \beta$$

$$F_y = F \cos \beta$$

Example 1: The forces **F1**, **F2**, and **F3**, all of which act on point **A** of the bracket, are specified in three different ways. Determine the **x** and **y** scalar components of each of the three forces.

The scalar components of **F**₁,

$$F_{1x} = 600 \cos 35^\circ = 491 \text{ N}$$

$$F_{1y} = 600 \sin 35^\circ = 344 \text{ N}$$

The scalar components of **F**₂,

$$F_{2x} = -500\left(\frac{4}{5}\right) = -400 \text{ N}$$

$$F_{2y} = 500\left(\frac{3}{5}\right) = 300 \text{ N}$$

The scalar components of **F**₃

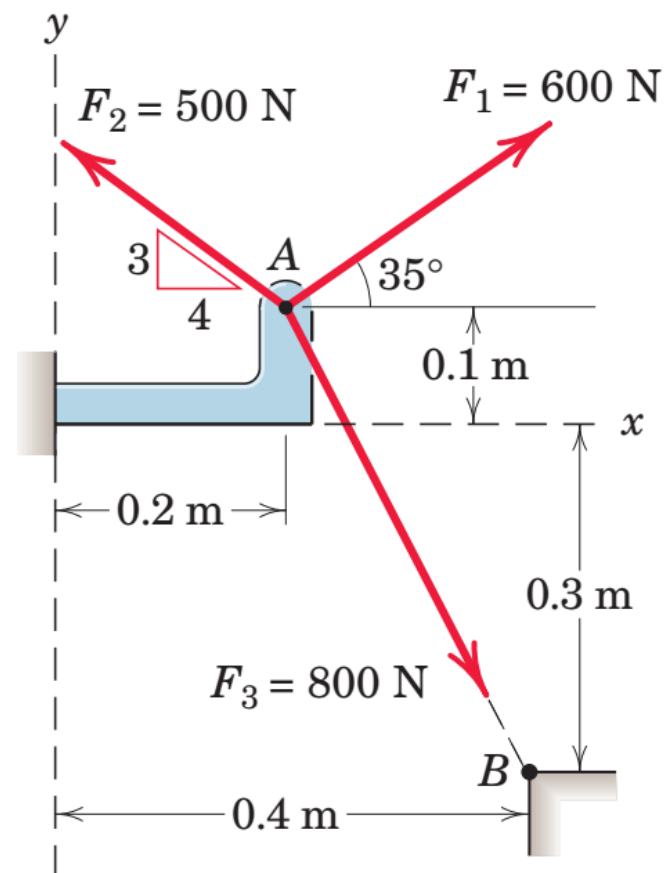
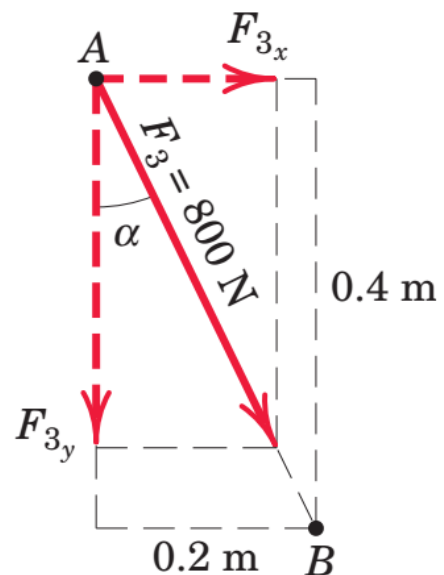
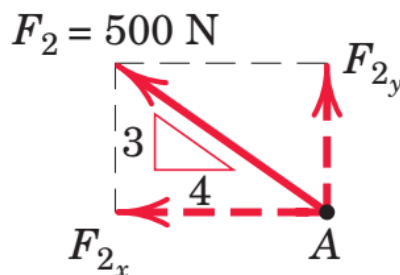
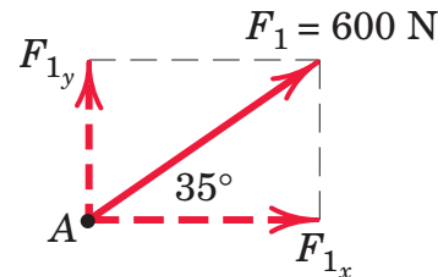
$$\alpha = \tan^{-1} \left[\frac{0.2}{0.4} \right] = 26.6^\circ$$

$$F_{3x} = F_3 \sin \alpha$$

$$= 800 \sin 26.6^\circ = 358 \text{ N}$$

$$F_{3y} = -F_3 \cos \alpha$$

$$= -800 \cos 26.6^\circ = -716 \text{ N}$$



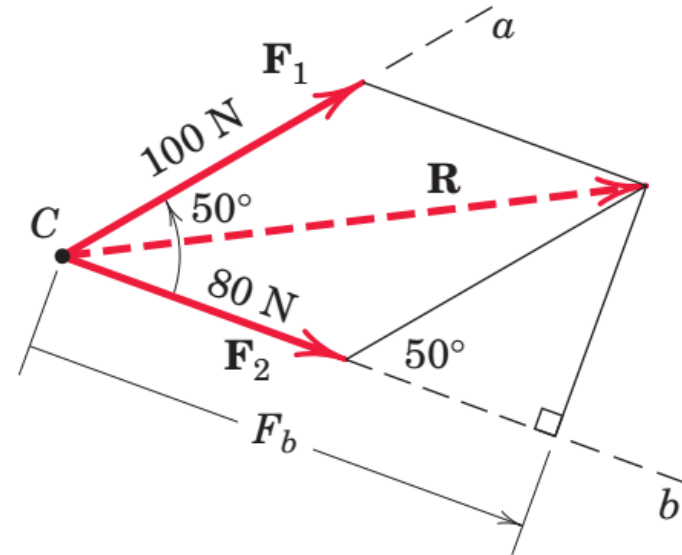
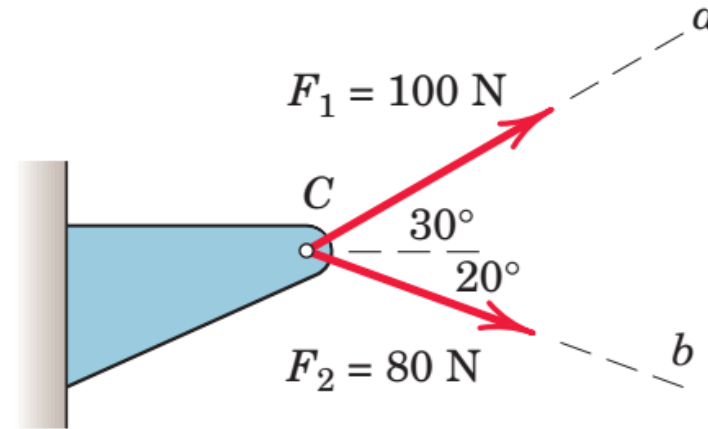
Example 2: Forces **F1** and **F2** act on the bracket as shown. Determine the projection **Fb** of their resultant **R** onto the **b-axis**.

Using the law of cosines gives us

$$R^2 = (80)^2 + (100)^2 - 2(80)(100) \cos 130^\circ$$

$$R = 163.4 \text{ N}$$

$$F_b = 80 + 100 \cos 50^\circ = 144.3 \text{ N}$$

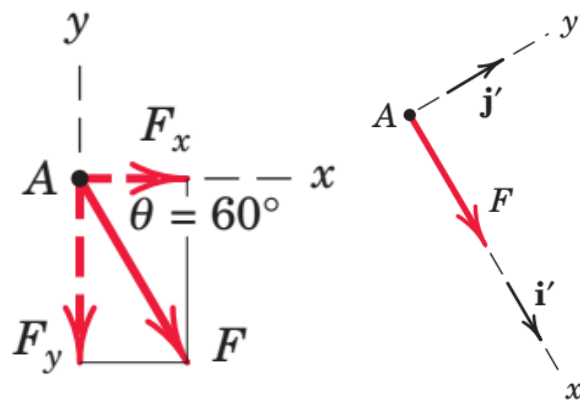


Example 3: The 500N force $\mathbf{F}$ is applied to the vertical pole as shown. (1) Write $\mathbf{F}$ in terms of unit vectors $\mathbf{i}$ and $\mathbf{j}$ and identify both its vector and scalar components. (2) Determine the scalar components of the force vector $\mathbf{F}$ along the x' and y' axes. (3) Determine the scalar components of $\mathbf{F}$ along the x and y' axes.

Part (1). $\mathbf{F} = (F \cos \theta)\mathbf{i} - (F \sin \theta)\mathbf{j}$

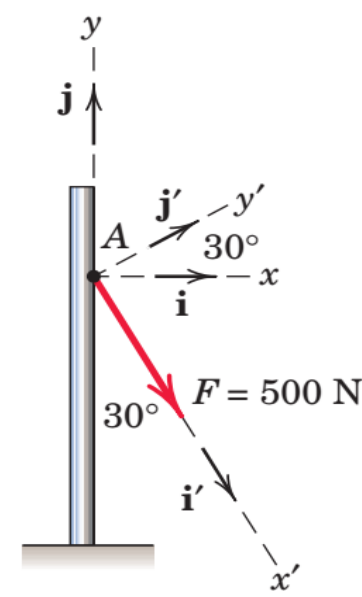
$$= (500 \cos 60^\circ)\mathbf{i} - (500 \sin 60^\circ)\mathbf{j}$$

$$\mathbf{F} = (250\mathbf{i} - 433\mathbf{j}) \text{ N}$$



The scalar components are $F_x = 250 \text{ N}$ and $F_y = -433 \text{ N}$.

The vector components are $\mathbf{F}_x = 250\mathbf{i} \text{ N}$ and $\mathbf{F}_y = -433\mathbf{j} \text{ N}$.



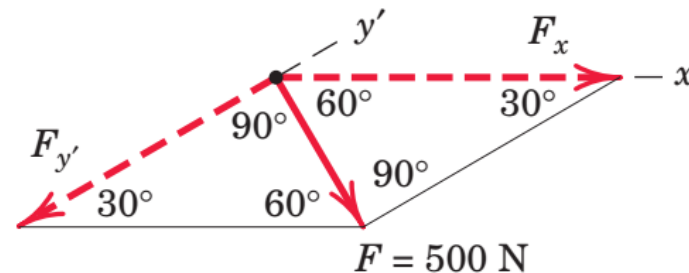
Part (2). we may write $\mathbf{F}$ as $\mathbf{F} = 500\mathbf{i}' \text{ N}$,

$$F_{x'} = 500 \text{ N} \quad F_{y'} = 0$$

Part (3). $\frac{|F_x|}{\sin 90^\circ} = \frac{500}{\sin 30^\circ} \quad |F_x| = 1000 \text{ N}$

$$\frac{|F_{y'}|}{\sin 60^\circ} = \frac{500}{\sin 30^\circ} \quad |F_{y'}| = 866 \text{ N}$$

$$F_x = 1000 \text{ N} \quad F_{y'} = -866 \text{ N}$$



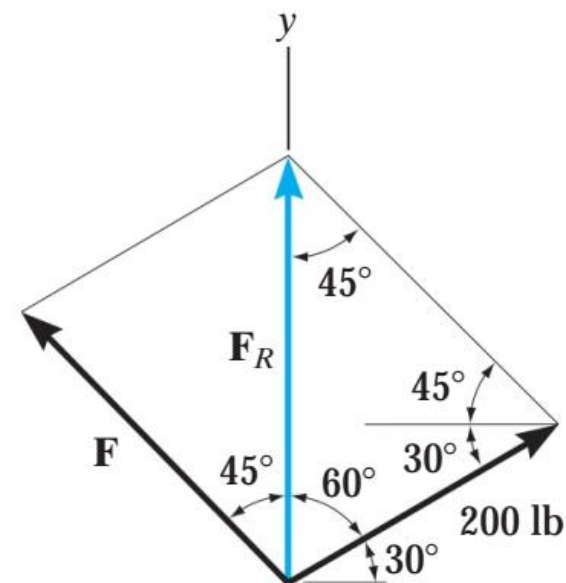
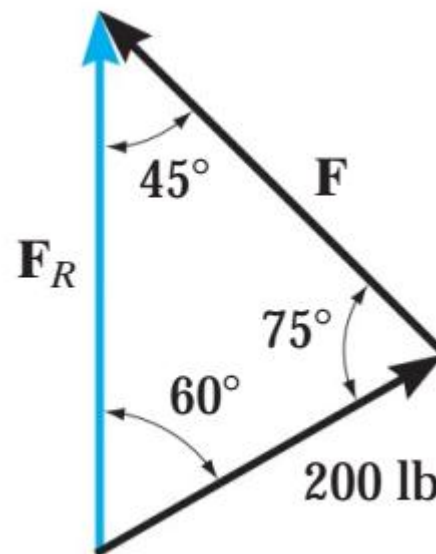
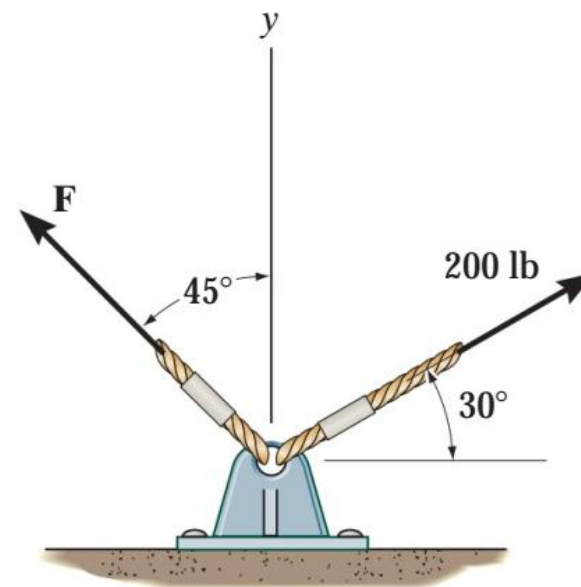
Example 4: Determine the magnitude of the component force F in Fig. and the magnitude of the resultant force $\mathbf{F}_R$ if $\mathbf{F}_R$ is directed along the positive Y axis.

$$\frac{F}{\sin 60^\circ} = \frac{200 \text{ lb}}{\sin 45^\circ}$$

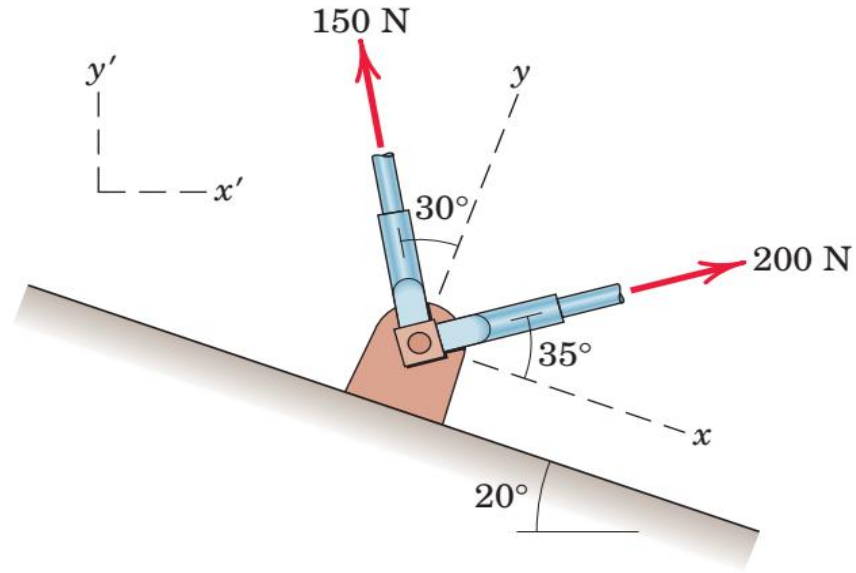
$$F = 245 \text{ lb}$$

$$\frac{F_R}{\sin 75^\circ} = \frac{200 \text{ lb}}{\sin 45^\circ}$$

$$F_R = 273 \text{ lb}$$



Example5 : Determine the resultant **R** of the two forces applied to the bracket. Write **R** in terms of unit vectors along the **x**- and **y**-axes shown.



Using the coordinates of the problem figure :

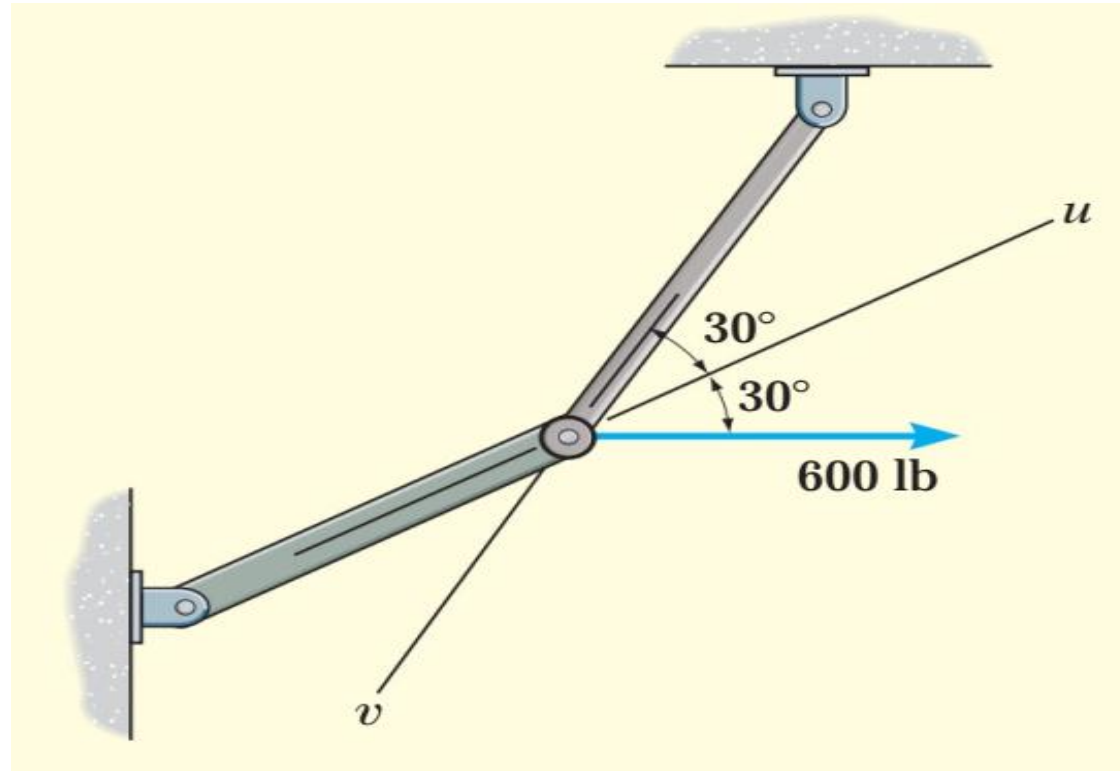
$$\begin{aligned} R_x = \sum F_x &= 200 \cos 35^\circ - 150 \sin 30^\circ \\ &= 88.8 \text{ N} \end{aligned}$$

$$\begin{aligned} R_y = \sum F_y &= 200 \sin 35^\circ + 150 \cos 30^\circ \\ &= 245 \text{ N} \end{aligned}$$

$$\therefore \underline{\underline{R = 88.8 \underline{i} + 245 \underline{j} \text{ N}}}$$

Homework-1

Resolve the horizontal **600lb** force in Fig. into components acting along the **u** and **v** axes and determine the magnitudes of these components.



Homework-2

The normal reaction force $\mathbf{N}$ and the tangential friction force $\mathbf{F}$ act on tire of a front wheel drive car as shown. Express the resultant $\mathbf{R}$ of these two forces in terms of unit vectors (a) $\mathbf{i}$ and $\mathbf{j}$ along the x - y axes, and (b) $\mathbf{e}_t$ and $\mathbf{e}_n$ along the n - t axes shown.

